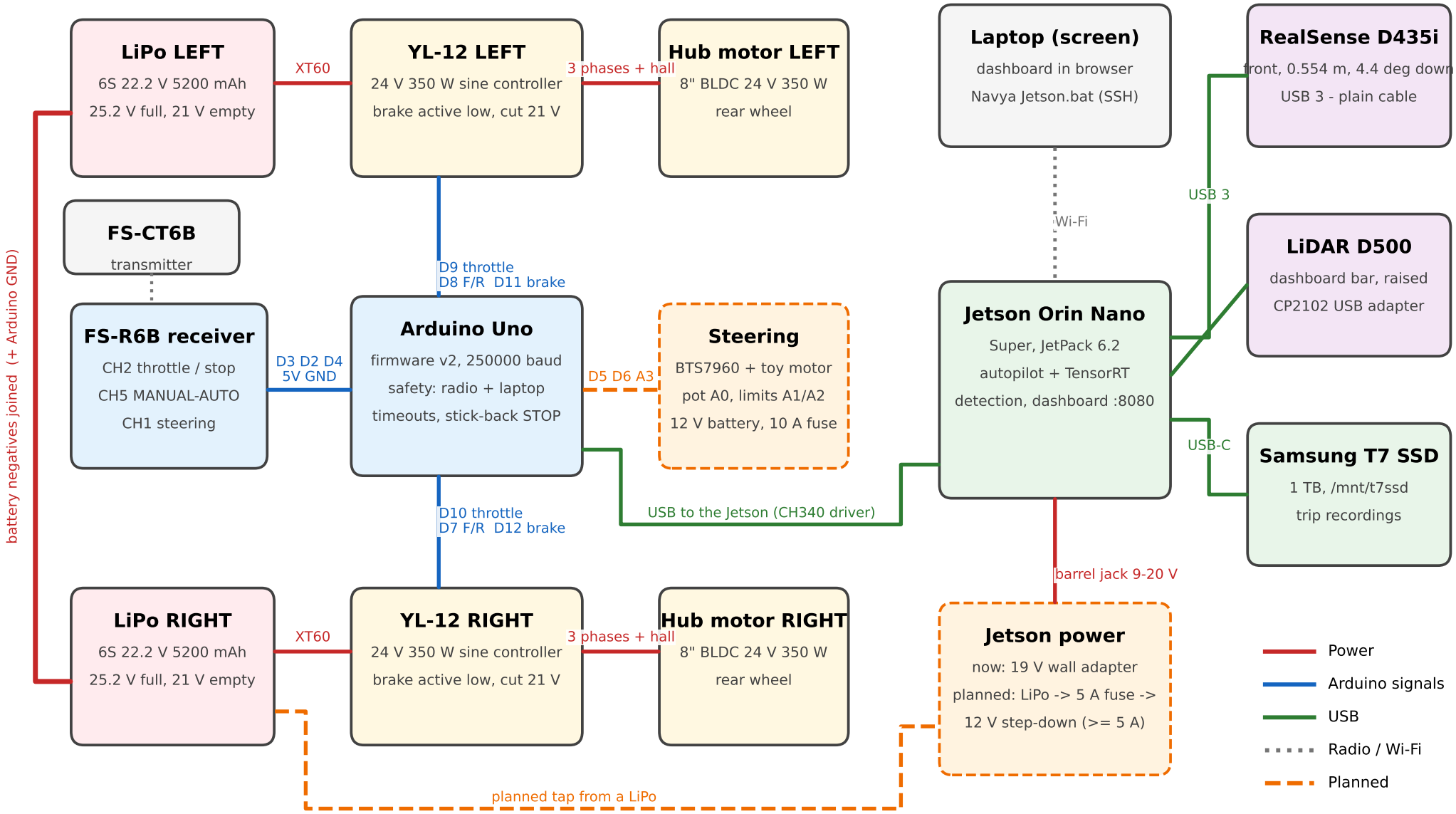
1. Whole car - how everything connects

Jetson Orin Nano drives the car; the laptop is only the screen (dashboard + SSH over Wi-Fi).



2. Arduino Uno - every pin

Firmware v2. Orange rows: wiring for parts still on order (steering, battery monitor).

Uno pin	Goes to
D9	LEFT controller throttle signal (purple)
D10	RIGHT controller throttle signal (purple)
D8	LEFT controller F/R (silver)
D7	RIGHT controller F/R (silver)
D11	LEFT controller brake wire (optional; active low)
D12	RIGHT controller brake wire (optional; active low)
GND	BOTH controllers' throttle black + F/R black (shared ground)
D3	Receiver CH2 signal (right stick up/down: throttle, pull back = STOP)
D2	Receiver CH5 signal (switch: MANUAL / AUTO)
D4	Receiver CH1 signal (right stick left/right: manual steering)
5V / GND	Receiver + / -
D5 / D6	BTS7960 RPWM / LPWM (steering - parts on order)
A3	BTS7960 R_EN + L_EN (steering)
A0	Steering potentiometer middle pin (outer pins 5V / GND)
A1 / A2	LEFT / RIGHT steering limit switch to GND
A4 / A5	Battery monitor: LiPo+ -47k- pin -10k- GND (left / right)
USB	Jetson USB-A port (or laptop) - 250000 baud, CH340 chip

Controller plug colours

Throttle: black = GND, orange = +5 V out (leave free), purple = signal. F/R: black = GND, silver = signal (open = forward). Brake: pull to GND = brake.

Must-haves

Uno GND joined to BOTH controllers' throttle GND - without it the throttle signal means nothing.
Battery negatives joined with a proper wire.

Radio

SWA DOWN (no dual rate). CH5 low = MANUAL, high = AUTO.
Right stick back (< 1300 us) = STOP + brake in any mode.
No radio for 0.1 s or no laptop/Jetson for 0.3 s = stop.

Recommended (cheap)

10 k from each purple throttle wire to GND, so a resetting Arduino reads as zero throttle.
Kill switch on both controllers' power-lock wires.

3. Jetson - ports, network and software

What goes into which Jetson port, and what to check if something is missing.

Device	Where	Notes
RealSense D435i camera	Jetson USB-A, straight in (USB 3, 5 Gbit/s)	Plain cable only. Through a USB-C dock it fails to start ('Port reset failed').
Arduino Uno (CH340)	Jetson USB-A	Needs the ch341 driver: bash jetson/build_ch341.sh
LiDAR D500 via CP2102 adapter	Jetson USB-A	230400 baud; found automatically
Samsung T7 SSD 1 TB	Jetson USB-C	Mounted at /mnt/t7ssd at start-up; trips in navya_trips
Power	Jetson DC barrel jack 5.5 x 2.5 mm	9-20 V, max 4.2 A. Now: 19 V wall adapter
Network	Wi-Fi	Same Wi-Fi as the laptop (phone hotspot / laptop hotspot)

Start a session (laptop) Double-click Navya Jetson.bat: finds the Jetson on the Wi-Fi, opens the menu on it, opens the dashboard when a drive starts.

Dashboard http://<jetson-ip>:8080 in the laptop's browser (the batch file opens it).

Pre-flight Menu option 1 - checks Arduino, radio, camera + calibration, LiDAR, SSD, detection, network.

Code on the Jetson ~/navya_autonomouscar, own Python in .venv (jetson/setup_jetson.sh); object detection = TensorRT engine (export_models.py).

Serial ports /dev/ttyUSB*: Arduino = CH340 (ch341 driver), LiDAR = CP2102 (cp210x). Found by USB ID.

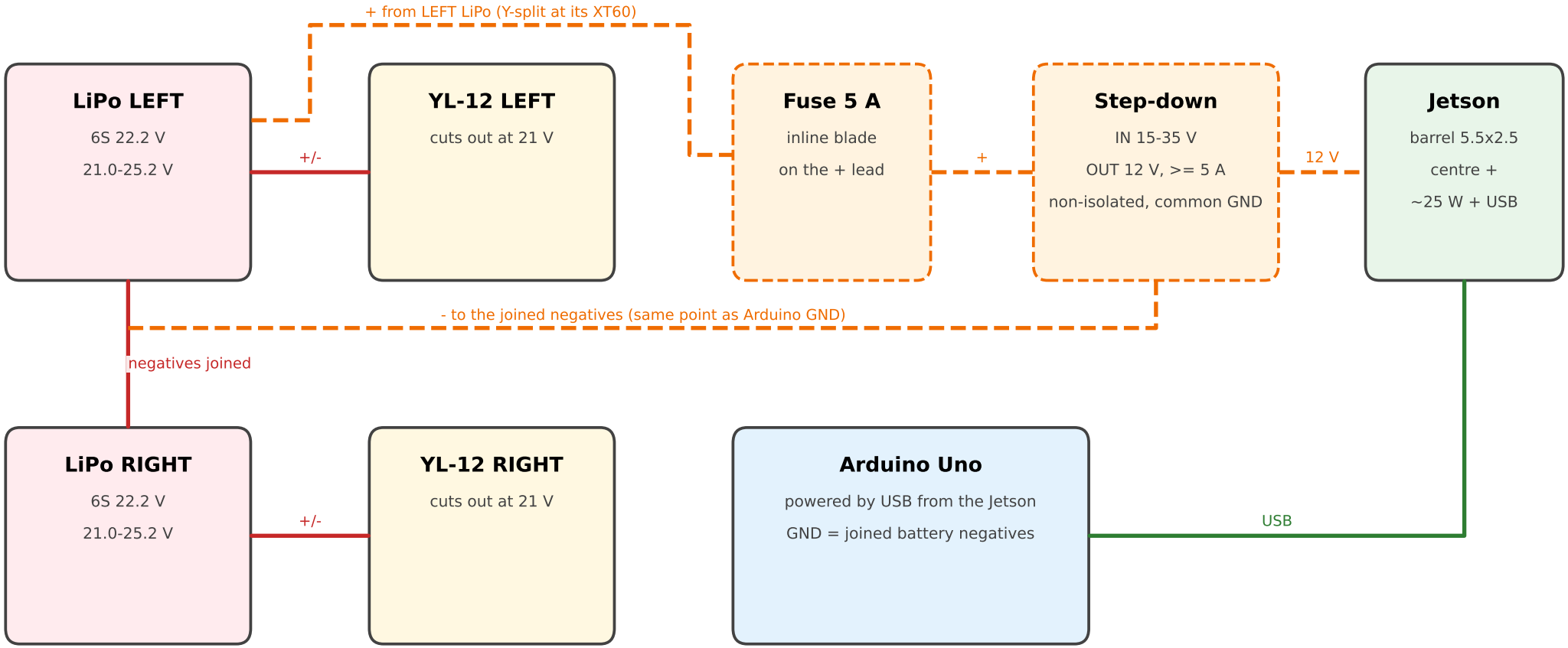
If the SSD is missing Plug it in before switching on, or run: sudo mount /mnt/t7ssd

If the camera is missing Plain cable straight into a Jetson port; unplug 10 s and re-plug; no docks or hubs.

If the Arduino is missing lsusb shows CH340 but no /dev/ttyUSB -> bash jetson/build_ch341.sh

4. Power - today, and the Jetson on the car (planned)

Jetson barrel jack accepts 9-20 V, max 4.2 A (19 V adapter supplied). A 6S LiPo gives 21-25.2 V.



1. Why 12 V and not 19 V: a 19 V converter needs a few volts of headroom, which a 6S pack near empty (21 V) does not have. 12 V sits safely inside the Jetson's 9-20 V range at every battery voltage.
2. Converter to buy: step-down (buck), input covering 21-26 V (e.g. 8-35 V), output 12 V fixed or adjustable, at least 5 A / 60 W. Set it to 12.0 V with a multimeter BEFORE connecting the Jetson.
3. Fuse 5 A on the LiPo + lead to the converter. Barrel plug 5.5 x 2.5 mm, centre positive (check before plugging in).
4. The LiPo keeps feeding the Jetson after the controllers cut out at 21 V: switch the Jetson off and unplug the converter after a session, or the pack is drained below 3 V per cell. The battery monitor (A4/A5) warns at 22.2 V.
5. Power-on order: Jetson first (it powers the Arduino over USB), then the controllers - they refuse to start if they see throttle at power-on. Unplug all LiPos when the car is not being tested.

5. On the car (stills from the walk-round video, 3 Nov 2026)

Replace with HD photos when convenient: docs/wiring/photos/, then run make_wiring_pdf.py again.

Controllers and steering wheel



Both YL-12 controllers behind the steering wheel, with their plugs.

RealSense camera (front)



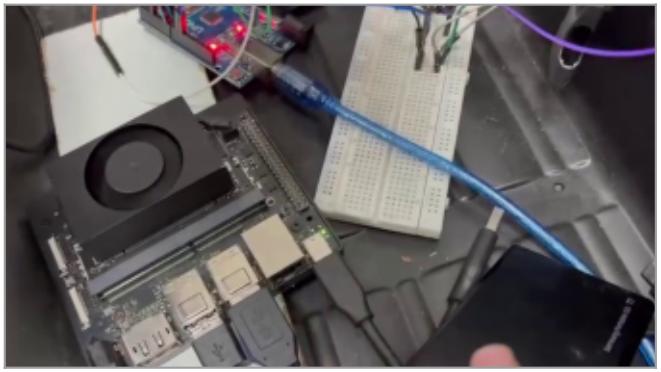
Front bonnet, centred, rigid bracket; 0.554 m high, 4.4 deg down. Toy steering motor below.

LiDAR (dashboard bar)



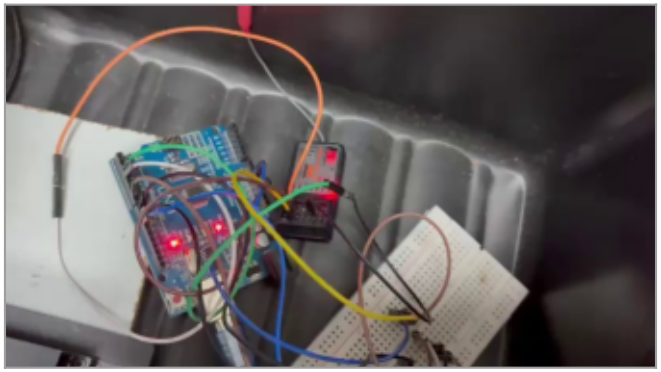
Raised above the camera, level; CP2102 adapter next to it. 0.365 m behind the bumper, 0.10 m left.

Jetson + SSD (footwell)



Jetson Orin Nano with the Samsung T7; Arduino (top) on the blue USB cable.

Arduino + receiver



Uno with jumper wires to the FS-R6B receiver and the breadboard rails (GND / 5 V).

Controller harness



YL-12 signal plugs (throttle, F/R, brake) and the joined grounds.

6. Before every run

Short version of the README bring-up. The car moves only with CH5 in AUTO.

1	**Car**	Wheels up on blocks for the first test of the day. Camera and LiDAR not knocked (pre-flight checks the camera).
2	**Transmitter**	ON, SWA down, CH5 MANUAL, right stick centred. One person holds it the whole run.
3	**Jetson**	SSD plugged in, then power on. Laptop and Jetson on the same Wi-Fi.
4	**Laptop**	Double-click Navya Jetson.bat. Option 1 = pre-flight: everything OK?
5	**LiPos**	Connect the controllers' LiPos last (Arduino already running).
6	**Drive**	Option 2 (or 3 with detection + recording). Dashboard opens by itself. CH5 to AUTO.
7	**Stop**	Right stick back = stop and brake, any mode. Q in the terminal ends the program.
8	**After**	CH5 MANUAL, unplug the LiPos, switch the Jetson off. Trips are on the SSD (navya_trips).